SMART INDIA HACKATHON 2026

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- Problem Statement ID –SIH26037
- Problem statement Title- Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- Theme - Smart Vehicles
- PS Category- Software
- Team ID-
- Team Name- Server Down

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INNOVATION AND UNIQUENESS

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IDEA/SOLUTION

Multi-sensor fusion of Camera + LiDAR + Radar for reliable perception in unstructured Indian road environments.

Short-term motion prediction for irregular and non-lane-based movement of vehicles, pedestrians and animals.

Lane-independent navigation for roads with missing, unclear or unreliable lane markings.

Designed specifically for diverse Indian mixed-traffic conditions rather than structured lane-based traffic.

Adaptive path planning that considers dynamic obstacles and their predicted trajectories.

Real-time replanning for sudden pedestrian movement, informal merging and unexpected obstacles.

EASE OF USE & ADAPTABILITY :

- Modular MATLAB and Simulink based simulation pipeline.
- RoadRunner-based realistic Indian road scenario creation.
- Continuous feedback from vehicle and environment enables dynamic replanning.
- Scenario-based testing allows repeatable validation under different traffic conditions.

CORE APPROACH :

Perception → Sensor Fusion → Motion Prediction → Adaptive Planning → Vehicle Control → Real-Time Replanning

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TECHNICAL APPROACH

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TECHNOLOGY

1. RoadRunner – Scenario Design
2. Automated Driving Toolbox – Sensor Simulation & Perception
3. Navigation Toolbox – Path Planning
4. Stateflow – Decision & Behaviour Logic
5. Simulink / Vehicle Dynamics Blockset – Vehicle Behaviour
6. Deep Learning Toolbox – Detection & Prediction

In 2025 alone, 513,563 road accidents and 183,382 deaths due to driver-related errors/bad driving and poor obstacle management (like potholes, due to wild animals), etc

Camera, LiDAR and Radar systems detects both static, moving and unexpected obstacles seperately

PERCEPTION

Combine observations from multiple sensors to estimate disturbances

SENSOR FUSION

Convert the planned trajectory into steering, acceleration and braking commands using a Simulink vehicle/bicycle model

VEHICLE CONTROL

MOTION PREDICTION

Predict short-term movement of surrounding agents to avoid irregular activities by vehicle owners

FEEDBACK LOOP

Vehicle movement → New sensor observations → Updated environment model → Replanning → Updated trajectory

ADAPTIVE PATH PLANNING

Generate a safe, collision-free path using dynamic obstacle positioning and predicted trajectories

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FEASIBILITY AND VIABILITY

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CHALLENGES

- Missing or unreliable lane markings on Indian roads.
- Diverse road users sharing the same road space.
- Sudden pedestrian and animal movement.
- Informal merging and unpredictable traffic behaviour.
- Unclear road boundaries and unexpected obstacles.
- Need for real-time replanning in changing environments.

PROPOSED SOLUTION

- Multi-sensor perception using Camera + LIDAR + Radar.
- Sensor fusion to improve environmental awareness.
- Object tracking and short-term prediction for proactive decision-making.
- Lane-independent planning for unmarked roads.
- Dynamic collision checking to trigger replanning.
- Vehicle-dynamics-aware planning to generate feasible trajectories.

VALIDATION

1. 5 realistic Indian road scenarios
2. MATLAB/Simulink simulation
3. Closed-loop testing
4. Performance evaluation

MATLAB SIMULINK

VALIDATION FLOW

RoadRunner Scenarios → MATLAB/ Simulink → Closed-Loop Testing → Performance Evaluation

Real-world Scenarios → Simulation → Test → Validate

SCALABILITY

- Modular architecture allows perception, prediction and planning modules to be improved independently.
- Additional road users and scenarios can be added without redesigning the complete system.
- Scenario-based simulation enables repeatable testing under different traffic conditions.

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IMPACT AND BENEFITS

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1 SAFETY

- Reduce collision risk in unpredictable mixed-traffic environments.
- Handle sudden pedestrian, animal and obstacle events.
- Enable safer navigation even without formal lane markings.

2 ADAPTABILITY

- Real-time path replanning according to changing surroundings.
- Handle irregular and non-lane-based road-user behaviour.
- Adapt to village roads, urban intersections, highways and dense markets.

3 RELIABILITY

- Multi-sensor perception for improved environmental awareness.
- Motion prediction enables proactive collision avoidance.
- Vehicle-dynamics-aware planning generates feasible trajectories.

4 PERFORMANCE

- Collision-Free Rate
- Scenario Completion Rate
- Replanning Latency
- Path Smoothness
- Minimum Safety Distance
- Scenario Completion Time

5 MACRO IMPACT

- Supports safer autonomous driving in diverse Indian road conditions.
- Provides a simulation-driven framework for testing autonomous vehicle behaviour.
- Enables systematic validation before real-world deployment.

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RESEARCH AND REFERENCES

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MATLAB related modules and terms

SNo.	LINK	DESCRIPTION
1	MathWorks – Automated Driving Toolbox	Sensor simulation, perception, tracking and automated driving workflows.
2	MathWorks – RoadRunner	Realistic road, traffic and autonomous driving scenario design.
3	MathWorks – Navigation Toolbox	Navigation and path planning algorithms.
4	MathWorks – Stateflow	Decision-making and autonomous vehicle behaviour logic.
5	MathWorks – Vehicle Dynamics Blockset	Vehicle dynamics and control simulation.
6	MathWorks – Deep Learning Toolbox	Object detection and trajectory prediction models.
7	MathWorks – Simulink	System modelling and closed-loop simulation.
8	MathWorks – RoadRunner Scenario	Scenario generation and autonomous driving testing.

ALL LINKS

SNo.	LINK	DESCRIPTION
1	MathWorks – Automated Driving Toolbox	Matrix Laboratory
2	MathWorks – RoadRunner	GitHub of all code
3	MathWorks – Navigation Toolbox	LIVE DEPLOYMENT

- MATLAB & Simulink simulation model
- RoadRunner-based Indian road scenarios
- Camera + LiDAR + Radar sensor simulation
- Closed-loop autonomous vehicle navigation
- Real-time adaptive path replanning

- Unmarked Village Road
- Busy Urban Intersection Without Signals
- Highway Merge with Slow-Moving Vehicles
- Dense Market Area with Mixed Traffic
- Sudden Cattle-Crossing Event

REQUIRED TEST SCENARIOS